

ESS Investigation

Space Systems

Unearthing Mars — A Historical Perspective

Teacher Materials

Investigation Overview

In this investigation, students will be assessed on their knowledge of and skills related to HS-ESS1-4. Students will plot the positions of Mars' orbit using the same process Johannes Kepler used to determine his laws of planetary motion. Johannes Kepler used Tycho Brahe's data to determine his laws of orbital motion. Brahe's data consisted of heliocentric longitudes of Earth (the position of Earth relative to the Sun) and geocentric longitudes of Mars (the position of Mars relative to Earth). Additionally, students will construct a model of the orbit of Mars using given data. These given data values and calculations will be compared to the student's own plots of Mars' orbit and will be used to determine percent error.

In **Session 1: Collaborative Student Work**, students will collaborate in pairs to individually construct two models of the orbit of Mars.

In **Part 1**, each student will:

- calculate the distance between foci, given the eccentricity and length of major axis values of Mars
- construct a model of the orbit of Mars and determine the aphelion, perihelion, and length of the major axis of Mars' orbit by using the student-calculated distance with string, pins, corkboard (or similar), calculators, and rulers

In **Part 2**, students will continue to collaborate in pairs to individually produce the orbit of Mars using the historical data sets provided. Each student will:

- plot four positions of Mars in its orbit using the heliocentric longitude of Earth and the geocentric longitude of Mars
- draw an estimated orbit for Mars by connecting the six plotted positions
- determine the aphelion and perihelion positions on the orbit of Mars
- measure the length of the major axis from the perihelion to the aphelion position
- determine the percent error of the calculated eccentricity (obtained from the student-drawn orbit from six points) to the given eccentricity value

In **Session 2: Individual Student Work**, students will independently answer the investigation questions in ***Student Answer Packet 2***.

Standards Alignment

Primary PE

HS-ESS1-4: Use mathematical or computational representations to predict the motion of orbiting objects in the solar system.

Related PEs

HS-ESS1-7: Construct an explanation using evidence to support the claim that the phases of the Moon, eclipses, tides, and seasons change cyclically.

HS-ETS1-3: Evaluate a solution to a complex, real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts.

Primary DCI

ESS1.B: Earth and the Solar System: Kepler’s laws describe common features of the motions of orbiting objects, including their elliptical paths around the Sun. Orbits may change due to the gravitational effects from, or collisions with, other objects in the solar system.

Related DCIs

ESS1.B: Earth and the Solar System: (NYSED) Earth and celestial phenomena can be described by principles of relative motion and perspective. (HS-ESS1-7)

ETS1.B: Developing Possible Solutions: When evaluating solutions, it is important to take into account a range of constraints, including cost, safety, reliability, and aesthetics, to consider social, cultural, and environmental impacts. (HS-ETS1-3)

Primary SEP

Using Mathematical and Computational Thinking: Use mathematical or computational representations of phenomena to describe explanations. Students will calculate the distance between foci using the given eccentricity and length of the major axis values and determine aphelion and perihelion positions. Students will then create one orbit of Mars using mathematical calculations, push pins, a loop of string, and a ruler. Students will create a second orbit of Mars from historical and modern data, using heliocentric longitudes of Earth and geocentric longitudes of Mars and a 10-cm circular protractor.

Related SEPs

(Constructing Explanations and) Designing Solutions: Evaluate a solution to a complex, real-world problem, based on scientific knowledge, student-generated sources of evidence, prioritized criteria, and trade-off considerations. (HS-ETS1-3) Students will evaluate the constraints engineers consider when designing Mars missions.

Constructing Explanations (and Designing Solutions): Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future. (HS-ESS1-7) Students will construct explanations and support claims with valid and reliable evidence from a variety of sources.

Primary CCC

Scale, Proportion, and Quantity: Algebraic thinking is used to examine scientific data and predict the effect of a change in one variable on another variable. Students will use algebraic thinking to calculate the distance between foci in the eccentricity equation. Students will also use angle relationships to locate the position of Mars in its orbit at different times throughout the Martian year.

Related CCCs

Influence of Science, Engineering, and Technology on Society and the Natural World: New technology can have deep impacts on society and the environment, including some that were not anticipated. Analysis of costs and benefits is a crucial aspect of decisions about technology. (HS-ETS1-3) Students will identify one benefit and one drawback of technology and engineering solutions for Mars missions.

Patterns: Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena. (HS-ESS1-7) Students will apply laws governing orbital motions of planets, moons, and human-made satellites to determine the eccentricity and location of Mars.

Teacher Directions

Preparing the Investigation:

This investigation, broken into **Session 1: Part 1**, **Session 1: Part 2**, and **Session 2**, is intended to satisfy a maximum of 140 minutes toward the 1200 minutes of hands-on laboratory experience required by part 100 of the Commissioner's Regulations. Investigations must be documented as satisfactory to be counted toward the 1200-minute requirement.

Appropriate safety practices are left to local discretion. The teachers should communicate the appropriate use and care of safety equipment and the associated procedures to students.

It is recommended that teachers model the triangulation method with students prior to Session 1. An additional student worksheet with two data sets and a practice student worksheet to plot two of Mars' positions have been provided at the end of this document and can be copied so teachers can model the process with students prior to the day of the investigation. A key has also been provided for your reference and for distribution to students at the end of the activity to check their work.

In **Session 1 (Part 1)**, students will collaborate in pairs to individually construct an orbit of Mars. Each student will use the formula for eccentricity to algebraically calculate the distance between foci, construct a model of the orbit of Mars using calculated values, and determine the aphelion and perihelion locations of the orbit.

In **Session 1 (Part 2)**, students will collaborate in pairs to individually plot locations of Mars using four historical data sets of heliocentric longitudes and geocentric longitudes. Each student will finish plotting the orbital positions of Mars, use the plots to draw the approximate orbit, and determine the aphelion and perihelion positions to determine the length of the major axis. Students will calculate the percent error of the calculated eccentricity (obtained from the student-drawn orbit from six points) to the given eccentricity value (found in ***Student Answer Packet 1.***)

In **Session 2**, students will individually complete ***Student Answer Packet 2.***

Determine how many station setups you will need. Although students will be working in pairs, each student will be responsible for completing ***Student Answer Packet 1*** individually. For example, if you have 30 students, you will need 30 setups.

Materials Needed for One Setup:

- 1 circular protractor (10-cm) (similar to the iplusmile 360° protractor, which has large numbers for students to read and continuous 0-360-degree markings)
- 1 ruler with centimeter scale
- 1 loop of string, 8.5 cm long when tied and stretched tight (total string length ~ 25-30 cm)
- 2 push pins
- 1 corkboard (12 in by 12 in) ½-in thick (or a similar medium such as foam board capable of holding push pins firmly in place)
- 4 different sharpened colored pencils used for plotting 4 different data sets (It is recommended that teachers have multiple individual sharpeners available for student use for precise line drawing during this investigation.)
- 1 calculator
- 1 pencil sharpener

Teacher Materials (not included with student setups):

- 1 class set of *Kepler's Triangulation Method Data for Practice Student Worksheet Model* and *Practice Student Worksheet*
- 1 projector/document camera for modeling the plotting of historical and modern Mars positions
- 1 circular protractor
- 1 ruler with centimeter scale

NOTE: It is recommended that transparent rulers and protractors are used for this investigation to assist students in making precise measurements and plotting points.

Preparation for Session 1: Part 1 of Investigation:

- 1) Assemble materials for the number of students in your class. (It is recommended that individual setups of student materials be compiled in individual, jumbo 2.5 gallon plastic bags, like Ziploc.)
- 2) Make sure *Kepler's Triangulation Method Practice Student Worksheets* (pages 11-13), ***Student Directions***, ***Student Answer Packet 1***, and ***Student Answer Packet 2*** have been copied and are ready for distribution to each student.
- 3) It is recommended that students complete the *Kepler's Triangulation Method Practice Student Worksheet* (page 12) prior to beginning **Part 1** of the investigation so that students have practice plotting the information in the data sets. Students will need to use the individual setups (materials in jumbo-sized bags) to complete this practice worksheet.
- 4) Assign pairs of students.
- 5) Distribute student materials (in jumbo-sized bags), ***Student Directions***, and ***Student Answer Packet 1***.
- 6) Prepare an area for all materials (jumbo-sized bags and student paperwork) to be stored at the end of **Session 1**.

Preparation for Session 1: Part 2 of Investigation:

- 1) Distribute all materials (jumbo-sized bags and student paperwork) back to each student.

Preparation for Session 2 of Investigation:

- 1) Make sure ***Student Answer Packet 2*** has been copied and is ready for distribution to each student.
- 2) Return ***Student Answer Packet 1*** and *Kepler's Triangulation Method Practice Student Worksheets* and the ***Student Directions*** to students, as they can be used for reference during **Session 2** in which the investigation questions are answered.

Student Directions with Teacher Annotations

Focus Questions: While performing this investigation, use the following focus questions to guide your work.

1. How did Kepler's Laws affect the ability of scientists to determine the locations and the motions of objects in our solar system?
2. Which mathematical and/or computational representations are used to plot the locations of Mars and predict its orbit?
3. How do gravitational effects change the orbits of objects in the solar system?
4. How can the principles of relative motion and perspective from Earth and other solar system objects be used to describe celestial phenomena today?

Inform students that they are not answering the focus questions at this time. The focus questions are included as an overview of concepts related to this investigation. These questions will be addressed through the students' performance of this investigation.

Procedure:

Session 1: Part 1 and Part 2

Session 1: Part 1 and **Session 1: Part 2** are each designed to be approximately 40 minutes in length. The break between **Part 1** and **Part 2** of **Session 1** will depend on the pace of student work (if completed during two different class periods). School districts with a block schedule should expect **Session 1: Part 1** and **Session 1: Part 2** to be completed during approximately one instructional block.

1. Write your name on ***Student Answer Packet 1***.
2. In ***Student Answer Packet 1***, the formula for eccentricity for Mars' orbit is missing one value, the distance between foci. The accepted value for Mars' eccentricity is 0.094. Use algebra to solve for this missing value. Show all work in the space provided. Round your answer *to the nearest tenth*.
3. Remove and place the *Model of Mars' Orbit Using Given Calculations* worksheet (page 3 of ***Student Answer Packet 1***) on the board. On this worksheet, plot the second focus position to the left of the "Sun" (using the distance calculated in step 2) with a large dot, and label this dot "Focus 2."

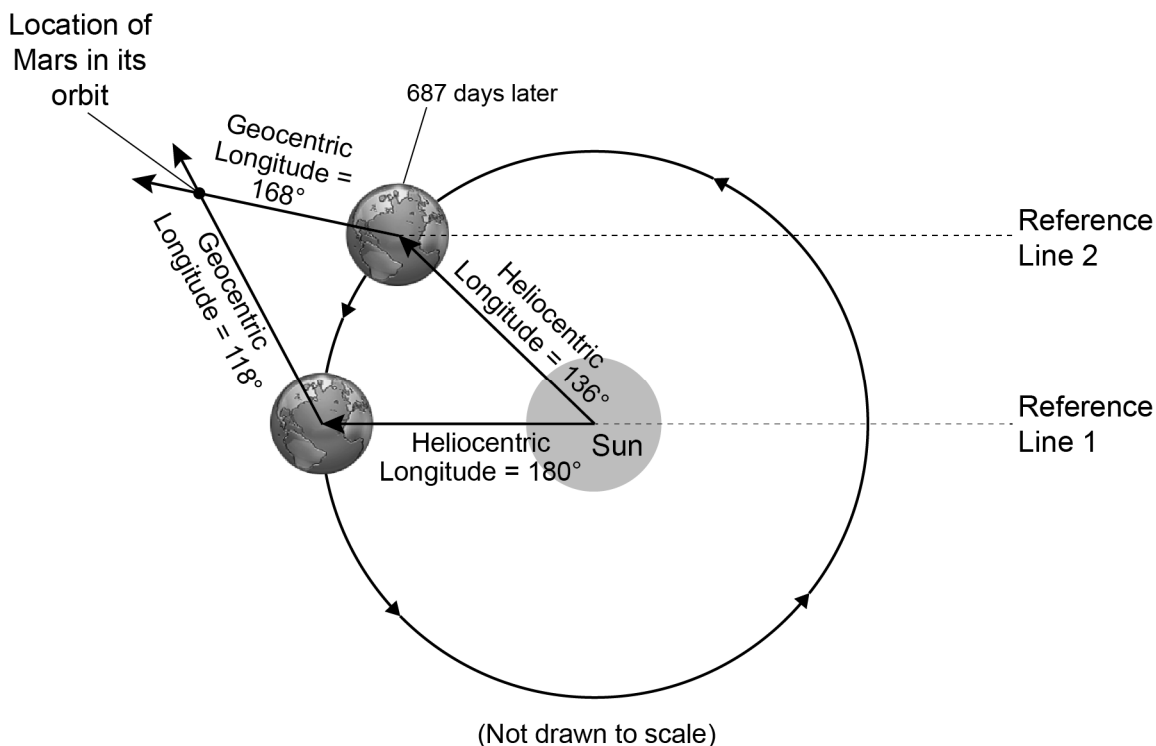
If the student uses the wrong worksheet, have additional copies available.

4. Place pins in each of the dots representing the Sun and the second focus.
5. Place the string *under* the plastic of the pins to construct an ellipse (be sure to stretch the string tight with a pencil). Label this ellipse "Mars' Orbit."

Instruct students to place the pins and string so that the string can be moved freely around the pins. This will prevent the string from being pinched beneath the pins.

6. Remove the pins from the paper.
7. Place one dot on your drawn “Mars’ Orbit” at the point closest to the “Sun’s” position. Label this dot “Perihelion.”
8. Place one dot on your drawn “Mars’ Orbit” at the point farthest from the “Sun’s” position. Label this dot “Aphelion.”

The model below illustrates the triangulation process for locating Mars by using two positions of Earth 687 Earth days apart. Two heliocentric longitudes show where Earth is relative to the Sun, and two geocentric longitudes are drawn and intersect to indicate the position of Mars outside of Earth’s orbit. Reference lines for measuring the heliocentric longitude and geocentric longitude are included.

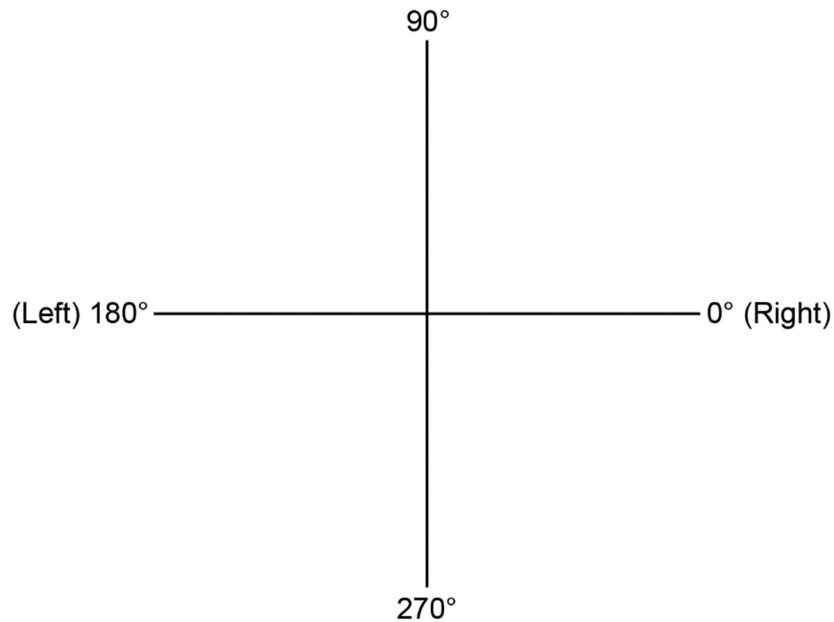


It is recommended that, prior to the first day of the investigation, the teacher demonstrate plotting Kepler’s data sets for two positions of Mars on the *Practice Student Worksheet Model of Mars’ Orbit Using Triangulation* by verbally modeling steps 9a through 9c below.

9. Remove the board. Using the *Data Table: Triangulation of Mars* on page 6, plot the four Mars Positions on the *Model of Mars’ Orbit Using Triangulation* on page 4 of **Student Answer Packet 1** by following the steps below (each Mars position will be triangulated using two dates 687 Earth days apart). The M1 and M2 Mars Positions (shaded in the data table) have already been plotted. You will plot Mars Positions M3 through M6.
 - a. The data in the “Heliocentric Longitude of Earth” column will be plotted on Earth’s orbit for Mars Position M3. Center the protractor on the Sun and align zero degrees with the Reference Line 1 (0°) Position. Plot the heliocentric longitude angle on Earth’s orbit with a dot using a colored pencil.

- b. The data in the “Geocentric Longitude of Mars” column will be plotted outside Earth’s orbit. Center the protractor on the plotted dot on Earth’s orbit and align zero degrees parallel to the horizontal lines of the grid.

The protractor must be centered on the Sun dot and match the numbers on Earth’s orbit.



Measure the geocentric longitude angle and draw a reference point outside Earth’s orbit to indicate this angle. Then draw a line using the same colored pencil approximately seven centimeters in length from the point on Earth’s orbit through the reference point outside Earth’s orbit.

Remind students that each of the four plots, M3 through M6, should be done in a different color. It is recommended that lighter shades (such as yellow) not be used, for the sake of visibility.

Remind students to align the numbers on the circular protractor with the numbers (0°, 90°, 180°, 270°) on the *Model of Mars’ Orbit Using Triangulation*.

Remind students that they should have all tasks (step 1 through step 16) completed by the end of **Session 1: Part 2**.

- c. Repeat steps 9a and 9b for the second date for Mars Position M4. The drawn geocentric longitude lines will intersect at a position outside Earth's orbit. The intersection of the two geocentric longitude lines locates the planet Mars for that pair of observations. Indicate this intersection with a dot, using one of the colored pencils, and label it M4.
- 10a. Repeat steps 9a, 9b, and 9c for Mars Positions M5 and M6. Be sure to place labels on all four Mars Positions. Each of the remaining Mars Position lines and dots should be a different color.
- 10b. Compare your four Mars positions with your partner.
11. Using your plotted points, draw the approximate orbit of Mars on your model so that the orbit passes through all six of the triangulated plotted Mars Positions. Label your orbit "Mars' Orbit."
12. Use a ruler to draw a line from Mars Position M1 to Mars Position M3 passing through the Sun. Label this line "Major Axis."
13. Measure the length of the Major Axis, in centimeters, *to the nearest tenth*. Record this value in **Student Answer Packet 1**.

It is recommended that students measure all distances between plots from center of plot to center of plot.

14. To find the distance between foci:
 - a. Measure the length from M1 to the Sun, in centimeters, *to the nearest tenth*. Record this value in **Student Answer Packet 1**.
 - b. Measure the length from the Sun to M3, in centimeters, *to the nearest tenth*. Record this value in **Student Answer Packet 1**.
 - c. Using your measured values from 14a and 14b, determine the Aphelion and Perihelion positions of the orbit. Label these positions "Aphelion" and "Perihelion" on either end of the Major Axis on page 4 of **Student Answer Packet 1**.
 - d. Subtract the Perihelion value (Sun to M3 distance) from the Aphelion value (M1 to Sun distance) to determine the distance between foci for this model. Record this value, in centimeters, *to the nearest tenth* in **Student Answer Packet 1**.
15. Calculate the eccentricity of Mars' orbit using the values from steps 13 and 14d. Show your work in the workspace and record this answer *to the nearest thousandth* in **Student Answer Packet 1**.
16. Calculate the percent error using your eccentricity value from step 15 and the accepted eccentricity for Mars' orbit found in step 2. Show your work in the workspace and record this answer *to the nearest tenth* in **Student Answer Packet 1**.
17. Place all your materials back into the jumbo-sized plastic bags.
18. Return all your materials and paperwork to your teacher.

Session 2

Return **Student Directions** and **Student Answer Packet 1** to the students as they can be used to answer some **Student Answer Packet 2** questions.

Your task in **Session 2** is to apply what you learned in **Session 1** to independently answer the investigation questions found in **Student Answer Packet 2**.

Data Table: Triangulation of Mars

Data	Mars Position	Date	Heliocentric Longitude of Earth (position of Earth relative to the Sun)	Geocentric Longitude of Mars (position of Mars relative to Earth)
Historical	M1	February 17, 1585	159°	135°
		March 15, 1587	115°	182°
	M2	March 28, 1587	196°	168°
		February 12, 1589	153°	218°
	M3	September 19, 1591	5°	284°
		August 6, 1593	323°	347°
	M4	December 7, 1593	86°	3°
		October 25, 1595	41°	49°
Modern	M5	May 26, 1935	245°	187°
		April 12, 1937	202°	246°
	M6	January 21, 1944	121°	66°
		December 8, 1945	76°	123°

Name: _____

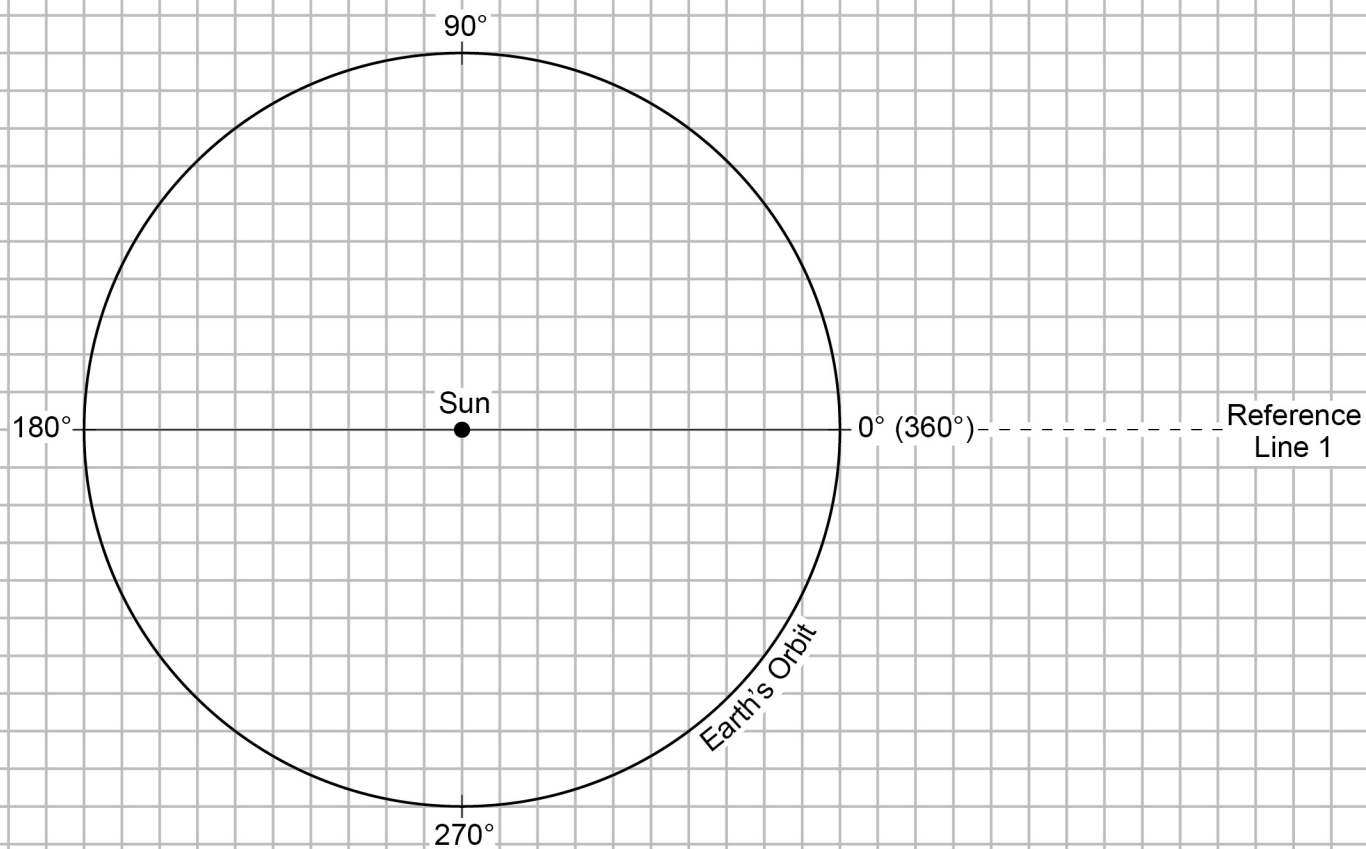
Kepler's Triangulation Method

Data for Practice Student Worksheet Model

Follow the teacher's instructions to practice plotting two of Kepler's data sets for two Mars Positions on the *Practice Student Worksheet Model of Mars' Orbit Using Triangulation*. Kepler used this data to determine the location of Mars in different positions in its orbit.

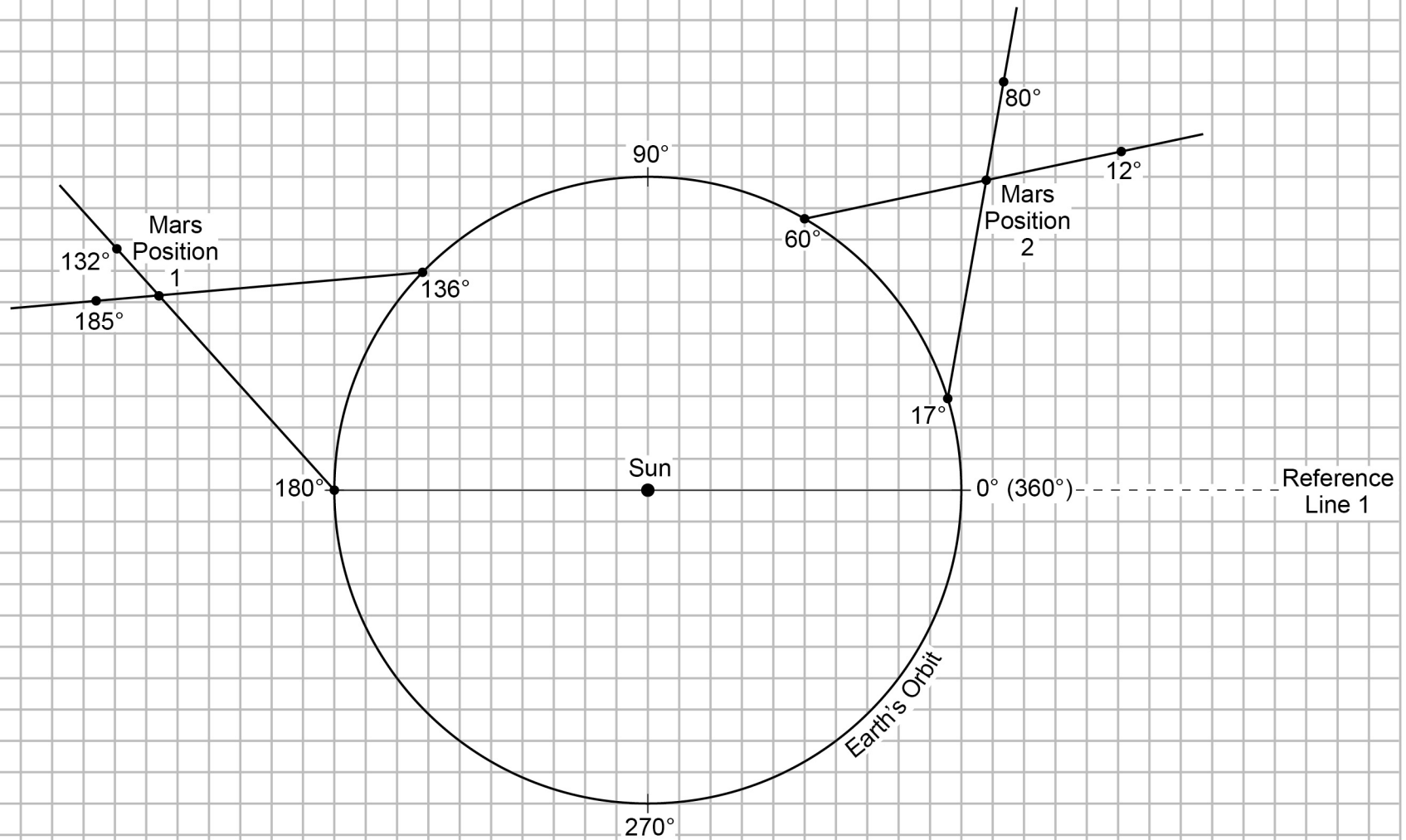
Date	Heliocentric Longitude of Earth	Geocentric Longitude of Mars	Mars Position
March 10, 1585	180°	132°	1
January 26, 1587	136°	185°	
November 22, 1941	60°	12°	2
October 10, 1943	17°	80°	

Name _____

Practice Student Worksheet: Model of Mars' Orbit Using Triangulation

Name _____

Practice Student Worksheet: Model of Mars' Orbit Using Triangulation (Answer Key)



Answer/Scoring Key**Student Answer Packet 1**2. Eccentricity of *Model of Mars' Orbit Using Given Calculations***Example of an acceptable response:**

$\text{Eccentricity} = \frac{\text{distance between foci (cm)}}{\text{length of major axis (cm)}}$ (The accepted value for Mars' eccentricity is 0.094. Source: NASA) Solve for x : $0.094 = \frac{x}{16.1 \text{ cm}}$
Workspace: An acceptable student response must include a correct mathematical calculation and answer. $(16.1) 0.094 = \frac{x}{16.1 \text{ cm}} (16.1)$ $1.5134 = x$
An acceptable student response must include a value correctly rounded to the nearest tenth of a cm. Distance between foci: <u>1.5</u> cm

13. Length of major axis: any value from 15.4 to 15.8 cm (15.6 cm +/- 0.2 cm)14. a. M1 to Sun: any value from 8.4 to 8.8 cm (8.6 cm +/- 0.2 cm)b. Sun to M3: any value from 6.8 to 7.2 cm (7.0 cm +/- 0.2 cm)d. Distance between foci: any value from 1.5 to 1.7 cm (using student values from 14a and 14b).

15. Eccentricity of *Model of Mars' Orbit Using Triangulation*: An acceptable student response must include a correct mathematical calculation and answer *to the nearest thousandth*.

Example of an acceptable response:

Workspace: $\text{Eccentricity} = \frac{\text{distance between foci (cm)}}{\text{length of major axis (cm)}}$ $\text{Eccentricity} = \frac{1.5 \text{ cm}}{15.6 \text{ cm}}$
Eccentricity = An acceptable value is 0.096 or any value between 0.085 and 0.109

16. Percent error calculation: An acceptable student response must include a correct mathematical calculation.

Example of an acceptable response:

Workspace: $\% \text{ Error} = \frac{\text{calculated value} - \text{accepted value}}{\text{accepted value}} \times 100$ $\% \text{ Error} = \frac{0.096 - 0.094}{0.094} \times 100$
% Error = An acceptable value of 2.1% <i>or</i> an acceptable value that includes a correct calculation using student values in #15 rounded <i>to the nearest tenth</i>.

Student Model of Mars' Orbit Using Calculations:

Acceptable models must include correct:

- plotting of the second focus position and labeling it “Focus 2”
- constructing an ellipse with two foci and labeling it “Mars’ Orbit”
- identifying and labeling “Perihelion”
- identifying and labeling “Aphelion”

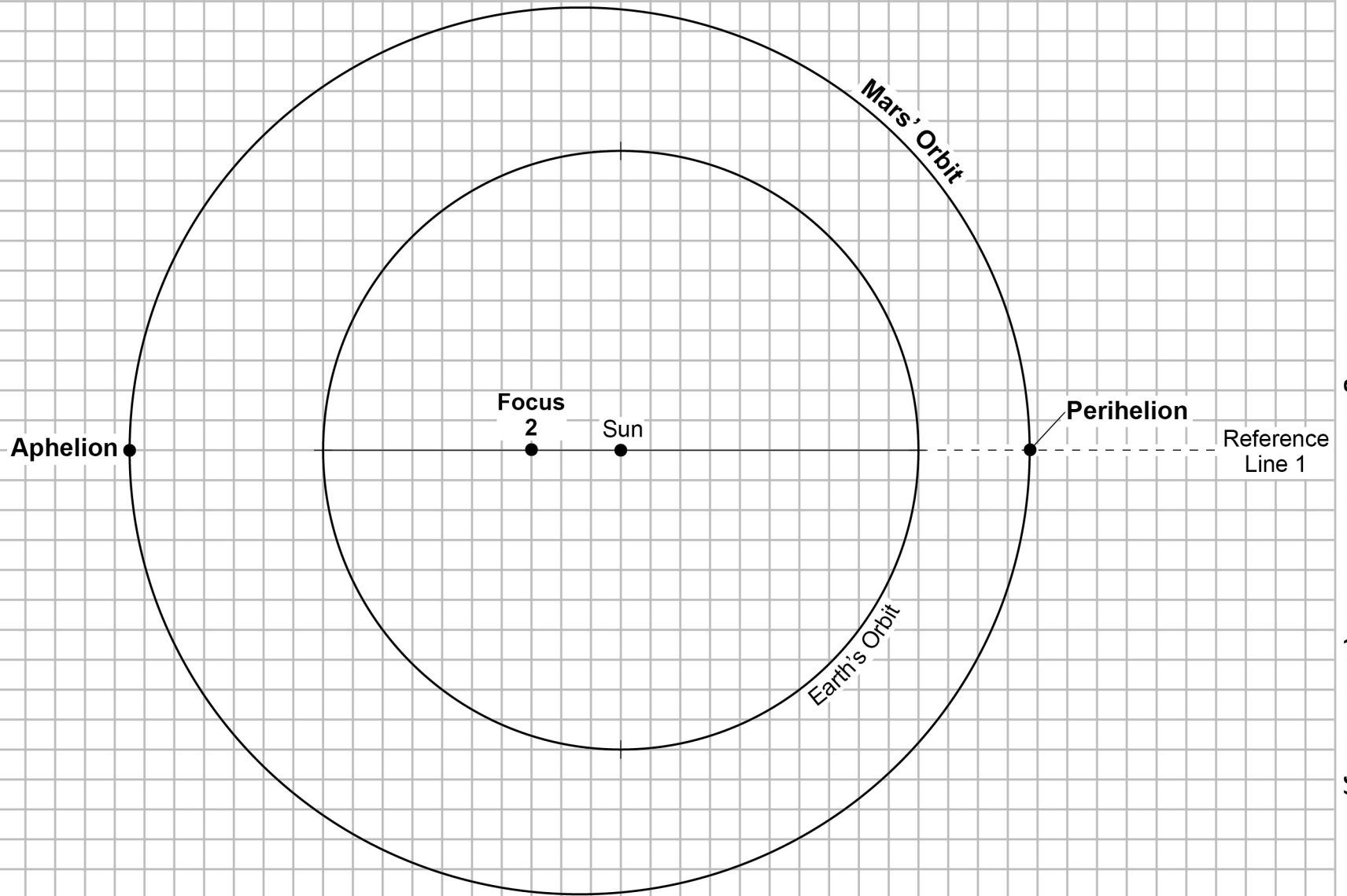
Student Model of Mars' Orbit Using Triangulation:

Acceptable models must include correct:

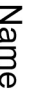
- plotting of the four positions of Mars
- measuring and drawing of geocentric longitude angles
- points of intersection of two geocentric longitude lines with a colored pencil and labeling of these points
- drawing and labeling of the “Major Axis”
- drawing of the approximate orbit of Mars through the plotted points
- identifying and labeling of “Perihelion”
- identifying and labeling of “Aphelion”

Name _____

Model of Mars' Orbit Using Given Calculations (Answer Key)



Name _____



Answer/Scoring Key**Student Answer Packet 2**

1 Acceptable responses include, but are not limited to:

- As distance from the Sun increases from 0.5 to 4.5 billion km, the planet's period of revolution increases from about 2000 to 60,000 days.
- The farther a planet is from the Sun, the longer it takes to revolve around the Sun. For example, at a distance of 1 billion km the number of days is approximately 7000, while at 2 billion km the number of days is approximately 1800.

2 Acceptable responses include, but are not limited to:

Law 1:

- Because of Kepler's First Law, scientists can determine the elliptical orbits of the planets.
- Orbiting bodies around the Sun have elliptical orbits.

Law 2:

- Scientists could determine the changing motion of the planets because if a line is drawn between the Sun and the planet, the planet sweeps equal areas in equal times, so the speed of the planet increases closest to the Sun and decreases when it is farthest from the Sun.
- Scientists can determine the velocity of planets based on their changing distance from the Sun at perihelion and aphelion.

Law 3:

- As the mean distance between the Sun and a planet increases, the period of revolution increases.
- According to this law, scientists can determine how long a planet would take to revolve around the Sun.

Note: Laws 1, 2, and 3 do not have to denote Kepler's 1st, 2nd, and 3rd laws.

3 Acceptable responses include, but are not limited to:

Orbit of Mars:

- To determine the orbit, we used the formula for eccentricity.
- We used the formula “length of the major axis divided by distance between the foci” to get the eccentricity and so we could draw the orbit.
- We used triangulation to find Mars’ locations and used those locations to draw a predicted orbit.

Positions of Mars:

- To determine the location of Mars in its orbit, we used the Heliocentric Longitude of Earth and the Geocentric Longitude of Mars to find four different locations in the orbit.
- We plotted the position of Earth, relative to the Sun, and the position of Mars, relative to Earth, to find four different locations of Mars using the triangulation method.

4 Acceptable responses include, but are not limited to:

- The perihelion and aphelion are opposite locations in a planet’s orbit, so they represent one half of the time it takes to complete the orbit. A year can be determined by multiplying this value by two.
- The time it takes a planet to travel from the perihelion to the aphelion points in an orbit are equal to one-half of a planet’s year. To determine the length of one year, double this value.

5 Acceptable responses include, but are not limited to:

- When a planet is at its perihelion point, it is closest to the Sun, so there is a greater gravitational attraction between the Sun and the planet. At aphelion, a planet is farthest from the Sun, so the gravitational attraction is less.
- Gravitational effects between the Sun and a planet are determined by the distance between them. At perihelion, the objects are the closest, so the attraction is the greatest. At aphelion, the two objects are farthest apart in their orbits, so the gravitational effects are the least.

6 Acceptable responses include, but are not limited to:

Model evidence:

- Location A represents the perihelion location, so the apparent diameter of the Sun would appear larger than at location B, which is farther away.
- At perihelion, the planet is closest to the Sun, so the model at location A is perihelion. At this location, the Sun's apparent diameter would appear larger than at location B.

Numerical evidence:

- At perihelion, the distance was about 7.0 cm to the Sun on the model of Mars' orbit, which was shorter than at aphelion, which was about 8.6 cm. This perihelion distance was represented as location A in the diagram, so the apparent diameter of the Sun would be larger than at location B, which was farther away.
- The apparent diameter of the Sun is greatest when the distance from the Sun to Mars is the smallest. This distance was about 7.0 cm on my model of Mars' orbit, which is the perihelion location.

Note: Acceptable numerical values must be consistent with those found in the student's *Student Answer Packet 1* (14a and 14b).

7 An acceptable response identifies gravity.

8 Acceptable responses include, but are not limited to:

- The velocity of the planet is different for A_1 , A_2 , and A_3 because at A_1 it is closer to the Sun and there is a greater gravitational force, while at A_3 it is farthest from the Sun, has a slower velocity, and less of a gravitational force between the Sun and Mars.
- The speed of the planet changes as it orbits the Sun because the gravitational force between Mars and the Sun changes.

9a&b An acceptable response must include a value from 250 to 260 Earth days *and* an acceptable explanation. Acceptable explanations include, but are not limited to:

- This path was selected instead of a straight-line path to Mars because Mars would not be in that location because it is moving every day in its orbit to a new location at a different speed than Earth travels in its orbit.
- The straight-line path from Earth to Mars does not account for the movement of Mars in its orbit or account for the time it takes the Curiosity Rover to travel that distance through space.

10 An acceptable response must include a correct description of speed for each satellite *and* a correct reason for this difference, using quantitative evidence. Acceptable responses include, but are not limited to:

Description of speeds:

- Satellite *A* moves faster than satellite *B*.
- Satellite *B* travels in its orbit slower than satellite *A*.

Reason with quantitative evidence:

- Satellite *A* moves faster than satellite *B* because it is only 200 km from Mars' surface, while satellite *B* is 500 km from Mars' surface. This difference in distance causes satellite *A* to be affected by a greater gravitational field/force.
- Satellite *B* will move slower than satellite *A* because it is 300 km farther away from Mars and affected by a weaker gravitational field/force.

11 Acceptable responses include, but are not limited to:

Solar panel benefit:

- The cost of solar panels to power a rover/lander is much cheaper than using radioactive batteries.
- The solar panels have a much smaller mass than the radioactive batteries, making the entire rover/lander lighter and more efficient.

Solar panel drawback:

- The solar panel does not work effectively, lasting for about half as long as the batteries.

Radioactive battery benefit:

- Batteries last a very long time and can prolong the life of the rover or lander for many years.

Radioactive battery drawback:

- The cost is much more expensive than using solar power.
- The mass of these power sources is much greater than solar power, which is a determining factor for the cost of a spacecraft to get the lander/rover to Mars.

12 Acceptable responses include, but are not limited to:

- Both locations have similar length of daylight hours and patterns of seasonal variations because both Mars and Earth have a similar degree of tilted axis and are at the same latitude.
- The patterns of length of daylight hours and seasons are similar for these locations because they are both located at the same latitude for the planet they are on and each planet has a similar degree of tilt to their axis of rotation.